

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

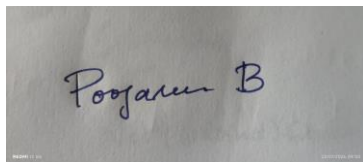
Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I POOJASREE B (192411091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer

Application	20	Correctly	Applies	Limited or	No or
of Knowledge		applies concepts to solve the question/problem	concepts with minor mistakes	partially correct application	incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Responsive Web Design (RWD)

RWD is a web development approach that enables a website to automatically adjust its layout, images, and content to fit different screen sizes such as mobile phone, tablets, and desktops.

Importance

- * Provides a better user experience on all devices.
- * Improves website accessibility and readability.
- * Reduces the need for separate mobile and desktop websites.
- * Improves SEO, as search engines

Key CSS Techniques

1. Flexible Layouts
2. Media Queries
3. Responsive Images
4. Relative Units

CSS Flexboxes or Grid

Flexboxes is suitable for 1D layouts
(rows or columns)

CSS Grid is suitable for 2D layouts
(rows and columns)

Example

HTML

```
<div class="container">  
  <div>Menu</div>  
  <div>Content </div>  
  <div>Sidebar </div>  
</div>
```

CSS

```
.container {  
  display: flex;  
  gap: 10px;  
}  
.container div {  
  flex: 1;  
  padding: 20px;  
  background: light blue;  
}  
@media (max-width: 768px) {  
  .container {  
    flex-direction: column;  
  }  
}
```

Improvements

- * Optimize images for faster loading
- * Use touch-friendly buttons
- * Ensure readable font sizes
- * Test the website on multiple screen sizes

2.

Client-side validation

Client-side validation is the process of checking user input in the browser the data is sent to the server. It is mainly performed using javascript.

Purpose

- * Prevents invalid data submission
- * Gives immediate feedback to users
- * Reduces unnecessary server requests
- * Improves user experience.

Javascript Event Handling

It uses submit, click and input to validate form field.

Example:

```
<form id="myform">
  <input type="email" id="email">
  <button type="submit">Submit </button>
</form>
<script>
document.getElementById("myform").addEventListener("submit",
function(e){
```

```

let email = document.getElementById("email").value;
let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

if (!pattern.test(email)) {
    alert("Invalid Email");
    preventDefault();
}

});
</script>

```

Role of RE

Role of Regular Expressions (Regex or RE) are patterns used to validate input formats.

Examples:

Email: `^[^\s@]+@[^\s@]`

Phn no: `^[8-9]{10}$`

Improvements:

- * Display error messages behind input fields instead of alerts.
- * Combine client-side validation with server-side validation for security.
- * Use HTML5 validation
- * Highlight invalid fields correctly.

3. CSS Box Model

The CSS Box Model is the fundamental concept that defines how every HTML element is displayed on a web page. Every element is treated as a rectangular box consisting of four components.

1. Content - The actual text, image or other content inside the element.
2. Padding - The space between the content and the border.
3. Border - The line that surrounds the padding and content.
4. Margin - The outer space outside the border that separates one element from another.

Understanding the box model helps developers control spacing, alignment and the overall layout of a webpage.

CSS Positioning Techniques

Static: Default positioning

Relative: Positioned relative to its normal position.

Absolute: Positioned relative to its nearest positioned ancestor.

Fixed: Stays fixed on the screen during scrolling.

Sticky: Switches between relative & fixed based on scrolling

Causes of Layout breaking

- * Fixed-width elements
- * Large images without responsive sizing
- * Improper positioning
- * Missing media queries
- * Overflowing content

Using Media Queries

```
• container {
```

```
  width: 80%;
```

```
}
```

```
@media (max-width: 768px) {
```

```
  • container {
```

```
    width: 100%;
```

```
  }
```

```
}
```

Best Practices

- * Use flexible layouts with flexbox or Grid
- * Use relative units (% , rem, vw)
- * Set images to max-width: 100%.
- * Apply box-sizing: border-box.
- * Test layouts on different devices and browsers.

4. Document Object Model (DOM)

It is a programming interface that represents an HTML document as a tree of objects. Javascript uses the DOM to access, modify, add or remove webpage elements dynamically without reloading the page.

Role of DOM

- * Access webpage elements
- * Change text, styles, and attributes
- * Add or remove HTML elements
- * Respond to user interactions.

Example:

```
< p id = " message " > Welcome! < / p >
```

```
< button onclick = " changeText () " > Click Here < / button >
```

```
< script >
```

```
function changeText () {
```

```
    document.getElementById ("message").innerHTML =  
        "Welcome to our website!";
```

```
}
```

```
< / script >
```

When the button is clicked, JavaScript updates the paragraph text instantly without refreshing the page.

Improvements

- * Use `textContent` for plain text.
- * Reduce unnecessary DOM updates.
- * Use `AJAX` or `Fetch API` for dynamic content.
- * Show loading indicators for a better user experience.